

Project 09

Inventory and Warehouse Analytics System (IWAS)

CSC 470: Software Engineering Lab

Department of Computer Science and Engineering

Project Specification Group Size: 4–5 Students Duration: One Semester

1 Project Overview

The Inventory and Warehouse Analytics System (IWAS) is a desktop or web-based application used by the warehouse manager of a trading company in Dhaka. The gate office already records every receipt into the warehouse (with quantity and purchase price), the dispatch desk already records every issue out of the warehouse, the purchase office already stores supplier delivery records, and departments already submit free-text purchase requisitions. The purpose of this software is *not* to add, delete or modify records, but to **analyse** the stored data, perform computations, match free-text requisitions to catalogue items using an AI-based text-matching technique, and print warehouse reports.

The system reads four input data sets:

1. **Item catalogue data:** item ID, name, unit, category, holding cost per unit per year, ordering cost per purchase order, and a short free-text description of the item.
2. **Stock movement data:** movement ID, item ID, date, type (receipt or issue), quantity, and unit purchase price (for receipts).
3. **Supplier data:** supplier ID, name, item ID supplied, promised delivery date and actual delivery date for each purchase order.
4. **Requisition data:** requisition ID, department, date, and the free-text description of the item requested (often imprecise, e.g. “black gel pens for the accounts section”).

2 Module Structure

The system consists of five modules. Each module has a distinct purpose and performs computational operations only.

No.	Module	Purpose
M1	Stock Valuation (FIFO)	Compute the value of closing stock using the first-in-first-out rule.
M2	Reorder and EOQ Analysis	Compute the economic order quantity and reorder point of every item.
M3	Supplier Performance	Compute on-time delivery percentage and average delay per supplier.
M4	AI Requisition-to-Catalogue Matching	Match free-text requisitions to the correct catalogue items.
M5	Reporting	Generate and print all warehouse reports.

3 Module M1: Stock Valuation (FIFO)

3.1 Purpose

For any selected item and date, this module computes the quantity in stock and its money value using the first-in-first-out (FIFO) rule: the units that arrived first are assumed to be issued first, so the closing stock consists of the most recently purchased units.

3.2 Operations and Formulas

$$S = \sum_r q_r - \sum_u q_u \quad (1)$$

$$V_{FIFO} = \sum_{r \in \mathcal{R}} \tilde{q}_r p_r \quad (2)$$

where:

- S = closing stock quantity of the item;
- q_r = quantity of receipt r ; q_u = quantity of issue u ;
- V_{FIFO} = FIFO value of the closing stock;
- $\mathcal{R}$ = the set of the most recent receipts that together make up the closing stock (found by consuming the oldest receipts first);
- $\tilde{q}_r$ = the portion of receipt r still remaining in stock;
- p_r = unit purchase price of receipt r .

3.3 Worked Example

A4 paper (reams) had three receipts and two issues this month:

Date	Type	Qty	Unit Price (p_r)
02/07	Receipt	100	BDT 520
06/07	Receipt	80	BDT 535
07/07	Issue	120	–
08/07	Receipt	60	BDT 550
08/07	Issue	30	–

Total receipts = 240, total issues = 150, so $S = 90$ reams. FIFO consumes the 02/07 receipt fully (100) and 50 of the 06/07 receipt, so the closing 90 reams consist of $\tilde{q} = 30$ reams at BDT 535 and 60 reams at BDT 550:

$$V_{FIFO} = 30(535) + 60(550) = 16050 + 33000 = \text{BDT } 49,050$$

3.4 Input Form: Valuation Enquiry Form

Field	Sample Input
Item	A4 Paper (ream)
As-of date	08/07/2026
Compute Valuation Print Report	
Output: Closing stock	90 reams
Output: FIFO value	BDT 49,050
Output: Layers shown	30 @ 535, 60 @ 550

4 Module M2: Reorder and EOQ Analysis

4.1 Purpose

For every item, this module computes the most economical quantity to order at a time (EOQ), the reorder point at which a new purchase order should be raised, and flags items that need ordering today. It also detects dead stock.

4.2 Operations and Formulas

$$Q^* = \sqrt{\frac{2 D S_o}{H}} \quad (3)$$

$$\bar{d} = \frac{D}{365} \quad (4)$$

$$\text{ROP} = L \bar{d} + 0.2 L \bar{d} \quad (5)$$

$$\text{Reorder now} \iff S \leq \text{ROP} \quad (6)$$

$$\text{Dead stock} \iff \text{no issue of the item in the last 180 days} \quad (7)$$

where:

- Q^* = economic order quantity (EOQ) of the item;
- D = annual demand of the item in units;
- S_o = ordering cost per purchase order (paperwork, transport);
- H = holding cost per unit per year (storage, insurance);
- $\bar{d}$ = average daily demand;
- L = supplier lead time in days;
- ROP = reorder point (the 20% addition is the safety stock);
- S = current stock on hand.

4.3 Worked Example

A4 paper: annual demand $D = 3,650$ reams, ordering cost $S_o = \text{BDT } 800$, holding cost $H = \text{BDT } 40$ per ream per year, lead time $L = 5$ days, current stock $S = 90$:

$$Q^* = \sqrt{\frac{2 \times 3650 \times 800}{40}} = \sqrt{146000} \approx 382 \text{ reams}$$

$$\bar{d} = \frac{3650}{365} = 10 \text{ reams/day}, \quad \text{ROP} = 5(10) + 0.2(5)(10) = 50 + 10 = 60$$

Since $S = 90 > 60$, no order is needed today; the module projects the order date as three days ahead, when stock will reach the ROP at the current usage rate.

5 Module M3: Supplier Performance

5.1 Purpose

This module measures how reliably each supplier delivers: the percentage of purchase orders delivered on or before the promised date, and the average delay of the late ones, so the purchase office can choose suppliers on evidence.

5.2 Operations and Formulas

$$\text{OnTime}_v (\%) = \frac{n_v^{on}}{n_v} \times 100 \quad (8)$$

$$\bar{\delta}_v = \frac{1}{n_v^{late}} \sum_{o \in late} (a_o - h_o) \quad (9)$$

where:

- OnTime_v = on-time delivery percentage of supplier v ;
- n_v = total purchase orders delivered by supplier v in the period;
- n_v^{on} = orders delivered on or before the promised date;
- n_v^{late} = orders delivered late ($n_v - n_v^{on}$);
- $\bar{\delta}_v$ = average delay of the late orders, in days;
- a_o = actual delivery date of order o ; h_o = promised date.

5.3 Worked Example

Supplier SUP-07 (Meghna Traders) delivered $n_v = 20$ orders this half-year, of which $n_v^{on} = 17$ were on time. The three late orders were delayed by 2, 4 and 6 days:

$$\text{OnTime}_v = \frac{17}{20} \times 100 = 85\%, \quad \bar{\delta}_v = \frac{2 + 4 + 6}{3} = 4 \text{ days}$$

Suppliers below 80% on-time are placed on the *watch list*; SUP-07 at 85% remains in good standing but its 4-day average delay is added to the lead time L used by Module M2 for its items — an example of two modules sharing a computed value.

6 Module M4: AI Requisition-to-Catalogue Matching

6.1 Purpose

Departments write requisitions in imprecise free text. This AI-based module matches each requisition against the description of every catalogue item and identifies what is actually being requested. An item whose description similarity with the requisition text is **at least 80%** is matched automatically and the requisition is priced from the catalogue; otherwise the requisition goes to the storekeeper's clarification queue.

6.2 Operation: Text Similarity by Cosine Measure

The requisition text and each item description are converted into binary term vectors A and C after removing stop-words:

$$\text{sim}(A, C) = \frac{A \cdot C}{\|A\| \|C\|} \quad (10)$$

where:

- A = binary term vector of the requisition text;
- C = binary term vector of the catalogue item description;
- $A \cdot C$ = dot product = number of terms common to both texts;
- $\|A\|, \|C\|$ = Euclidean norms, i.e. the square roots of the numbers of distinct terms in A and C respectively.

Groups may optionally upgrade the binary vectors to TF-IDF weights: $w_t = tf_t \times \log \frac{N}{df_t}$, where tf_t is the frequency of term t in the text, N is the total number of item descriptions, and df_t is the number of descriptions containing term t .

6.3 Worked Example

Catalogue item IT-3320 description terms: $C = \{pen, gel, black, ink, office\}$, so $\|C\| = \sqrt{5}$.

Requisition RQ-0871 reads: “*Black gel pens with smooth ink for the accounts office, one dozen.*” After stop-word removal the distinct terms are $A = \{black, gel, pen, ink, smooth\}$, so $\|A\| = \sqrt{5}$.

Common terms = $\{black, gel, pen, ink\}$, hence $A \cdot C = 4$.

$$\text{sim}(A, C) = \frac{4}{\sqrt{5}\sqrt{5}} = \frac{4}{5} = 0.80 = 80\%$$

Since $80\% \geq 80\%$, requisition RQ-0871 is matched to **IT-3320 (Gel Pen, Black)** and priced from the catalogue. The same requisition scores only 18% against the “whiteboard marker” description, confirming the match. The quantity word (*one dozen*) is converted to 12 units by a small unit-parsing rule.

6.4 Input Form: Requisition Matching Form

Field	Sample Input
Requisition ID	RQ-0871
Requisition text (read-only)	Black gel pens with smooth ink for the accounts...
Run AI Matching	
Output: Matched item	IT-3320 — Gel Pen, Black (80%)
Output: Quantity parsed	12 units
Output: Catalogue price	BDT 15/unit, total BDT 180
Output: In stock?	Yes (430 units)

7 Module M5: Reporting

All reports are printable on A4 with the company header, report title, date and page number. Sample formats with values follow.

7.1 Report R1: Daily Stock Movement Report

Padma Trading Ltd — Daily Stock Movement Report					
Date: 08 July 2026					
Item	Unit	Opening	Received	Issued	Closing
A4 Paper	ream	60	60	30	90
Gel Pen (Black)	piece	442	0	12	430
Toner Cartridge	piece	25	10	4	31
Movements recorded today: 5 receipts, 9 issues					

7.2 Report R2: Stock Valuation Report (FIFO)

Padma Trading Ltd — Stock Valuation Report (FIFO), 08 July 2026 (BDT)				
Item	Closing Qty	FIFO Layers	Unit Prices	Value
A4 Paper	90	30 + 60	535 / 550	49,050
Gel Pen (Black)	430	430	15	6,450
Toner Cartridge	31	21 + 10	5,200 / 5,350	162,700
Total warehouse value				218,200

7.3 Report R3: Reorder List Report

Padma Trading Ltd — Reorder List, 08 July 2026					
Item	Stock (S)	ROP	EOQ (Q^*)	Days to ROP	Action
Toner Cartridge	31	34	45	0	Order 45 now
A4 Paper	90	60	382	3	Order in 3 days
Stapler	55	6	20	–	OK
Items to order today: 1 Purchase requisition draft printed for the purchase office.					

7.4 Report R4: Supplier Performance and Dead-Stock Report

Padma Trading Ltd — Supplier Performance, January–June 2026				
Supplier	Orders (n_v)	On Time	Avg. Delay ($\bar{\delta}_v$)	Standing
Meghna Traders	20	85%	4 days	Good
Jamuna Supplies	14	71%	7 days	Watch list
Surma Stationers	9	100%	–	Excellent
<i>Dead stock (no issue in 180 days)</i>				
Fax Machine Ribbon — 44 units, value BDT 8,800 — disposal recommendation printed.				

7.5 Report R5: AI Requisition Matching Log

Padma Trading Ltd — AI Requisition Matching Log, 08 July 2026				
Req. ID	Department	Matched Item	Similarity	Status
RQ-0871	Accounts	IT-3320 Gel Pen, Black	80%	Priced, in stock
RQ-0872	Admin	IT-1108 A4 Paper	92%	Priced, in stock
RQ-0873	IT	best score 51%	–	Clarification queue
Requisitions today: 3 Auto-matched: 2 Clarification: 1				

8 Deliverables and Assessment

1. Software Requirements Specification (use cases, functional and non-functional requirements) — 15%
2. Design documents: context diagram, DFD (level 0 and 1), ER diagram, class diagram — 20%
3. Working implementation of modules M1–M5 with the formulas above, including a correct FIFO layer computation — 35%
4. AI requisition-matching module with a demonstration on at least ten sample requisitions — 15%

5. Printed reports R1–R5 with real computed values, and final viva — 15%

Constraint reminder: the system must not implement add, delete or update operations; all marks are awarded for analytical computation, AI-based matching and reporting.